Queue Management System Using Zoho Creator

Digital Transformation of Customer Queue Management

Presented by: Saisanjay SS

Roll No: 240701458

Rajalakshmi Engineering College



Customers wait in a digital queue

Introduction

1

What is a Queue Management System?

A system that issues tokens and manages customers' waiting lines in order.

2

Why manage waiting lines?

It reduces waiting time and avoids confusion and crowding at counters.

3

What is Zoho Creator?

A low-code platform for building custom business applications quickly.

4

How does it help customers?

A digital queue gives faster, fairer and more transparent service.

Problem Statement & Existing System

Problem statement

- Long waiting times.
- Manual token management.
- Poor customer communication.
- Difficulty managing multiple service counters.
- Lack of real-time queue information.

Existing system

- Traditional manual queue management.
- Paper-based token generation.
- Manual customer calling.
- No centralized monitoring.
- Limitation: slow, error-prone and hard to track.

Proposed System & Objectives

Proposed system

- Automated token generation.
- Digital queue tracking.
- Staff and counter management.
- Real-time status updates.
- Centralized administrator dashboard.

Objectives

- Reduce customer waiting time.
- Automate token allocation.
- Improve service efficiency.
- Manage multiple counters.
- Maintain customer and service records.
- Generate queue performance reports.

Technologies Used

1

Zoho Creator

Low-code platform used to build the application.

2

Deluge Scripting

Automates token logic and status updates.

3

Creator Forms

Collect customer and service data.

4

Creator Reports

View and manage records.

5

Pages & Workflows

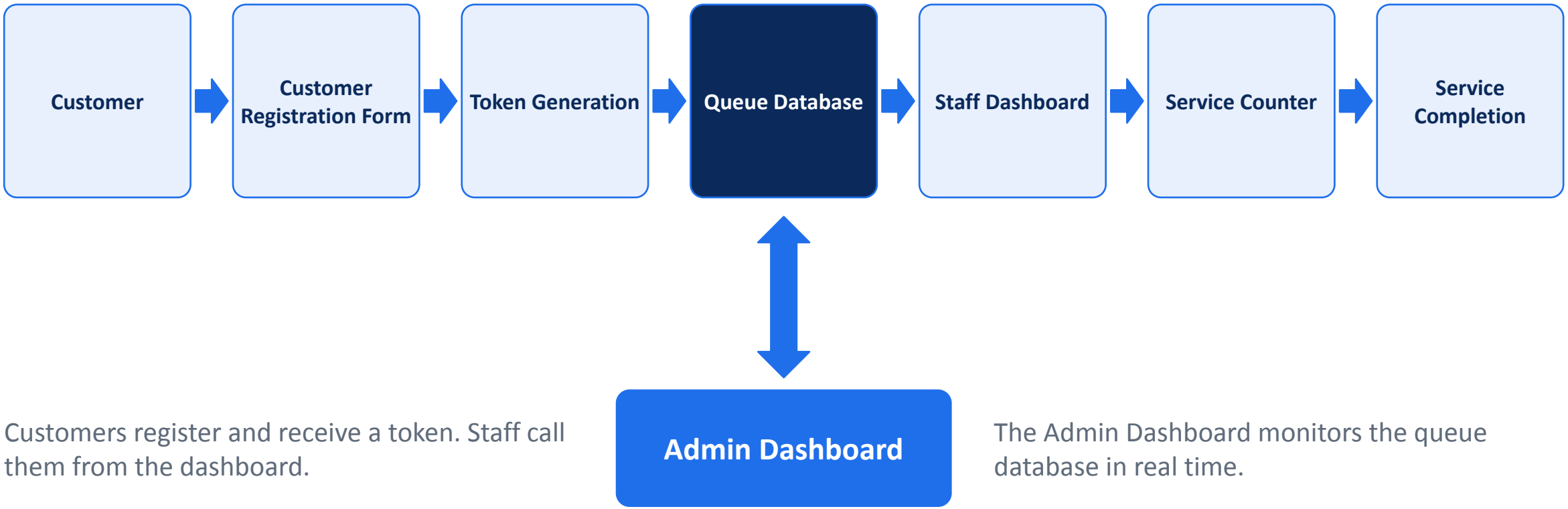
Build dashboards and automate actions.

6

Application Database

Stores tokens, customers, staff and counters.

System Architecture



Customers register and receive a token. Staff call them from the dashboard.

The Admin Dashboard monitors the queue database in real time.

Database Design

Customer Registration

- Customer ID
- Name
- Mobile Number

Queue Token

- Token Number
- Customer
- Service
- Date
- Status

Service Master

- Service Name
- Prefix
- Active Status

Counter Master

- Counter ID
- Counter Name
- Assigned Staff

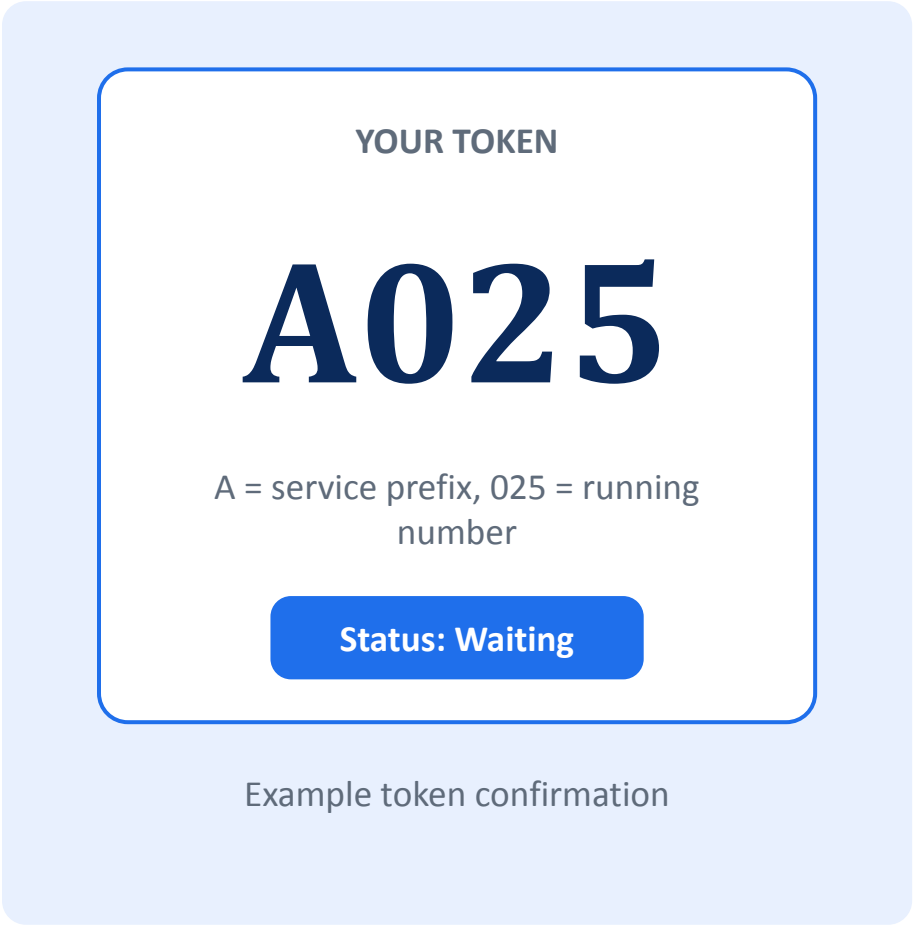
Staff Master

- Staff Name
- Email
- Assigned Counter

Queue Token is the central form: it links a Customer to a Service and tracks the Status.

Token Generation Module

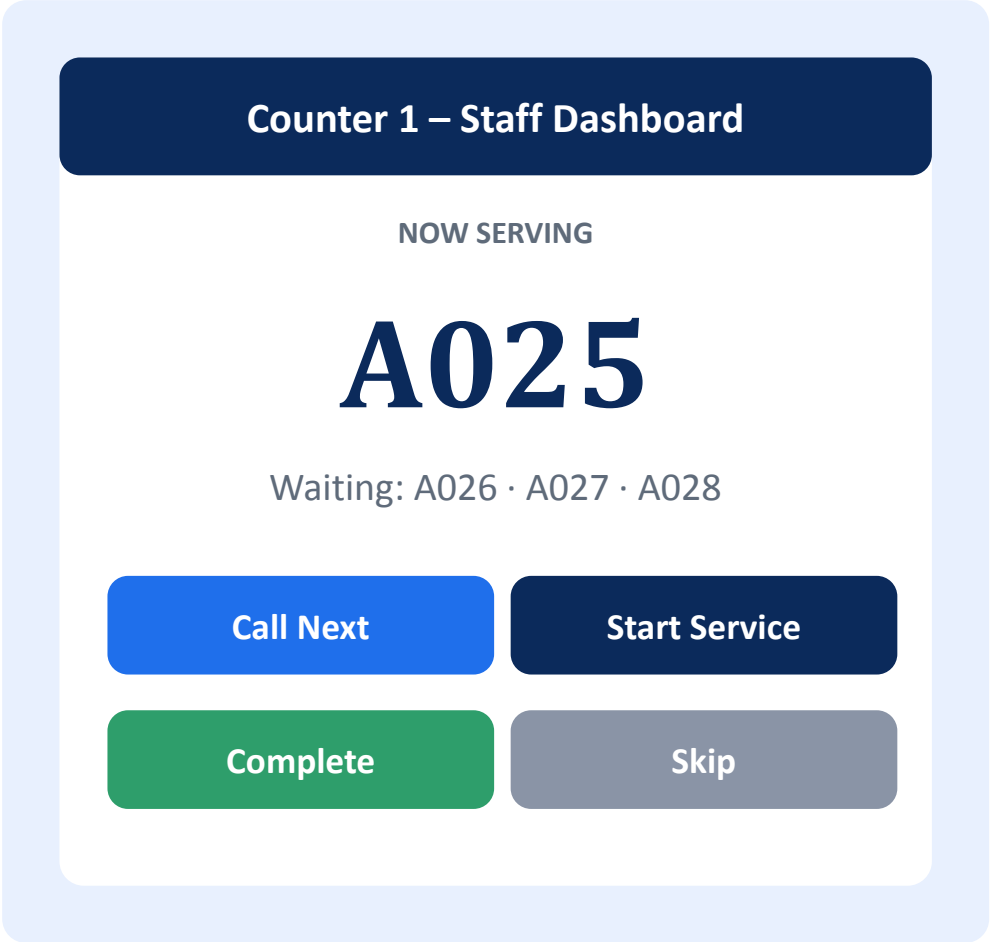
1. Customer enters registration details.
2. Selects the required service.
3. System generates a unique token number.
4. Token is saved in the database.
5. Customer receives token confirmation.



Example token confirmation

Staff & Counter Management

- Staff login.
- View waiting customers.
- Call the next available token.
- Start customer service.
- Complete or skip tokens.
- Manage assigned counters.



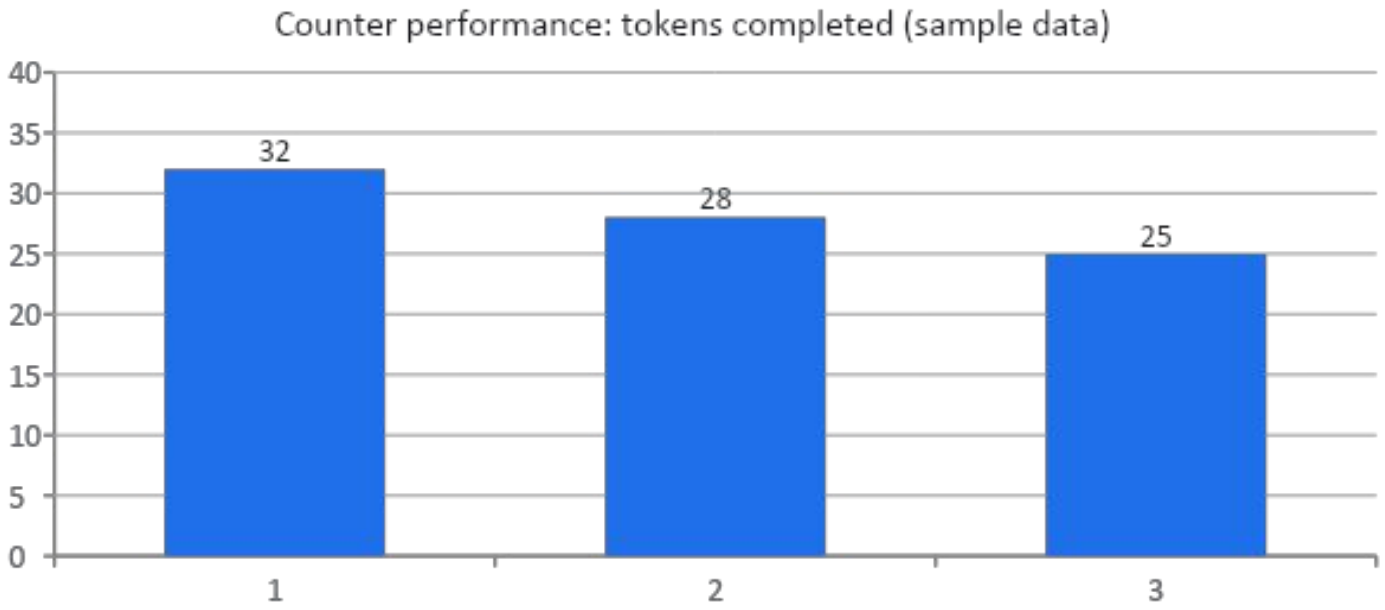
Admin Dashboard & Reports

100
Total tokens generated

12
Waiting customers

3
Currently serving

85
Completed customers

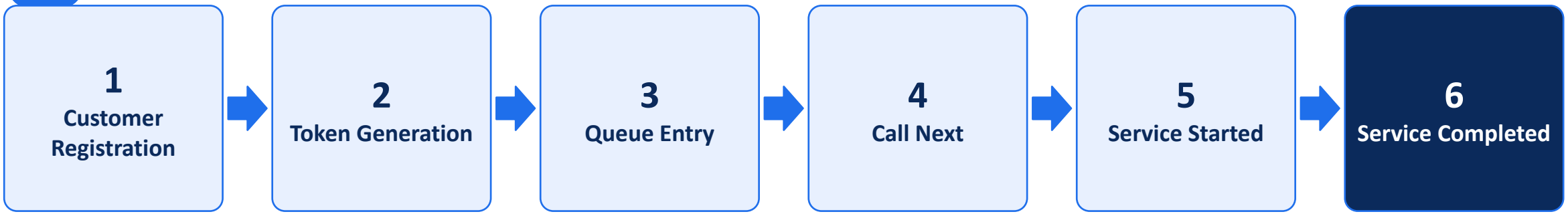


Reports

- Counter performance.
- Daily reports.
- Monthly reports.

Figures shown are sample data for illustration.

Workflow & Implementation



How Zoho Creator automates it

Workflows

Run automatically when a form is submitted or a status changes.

Deluge scripts

Generate token numbers such as A025 and update token status.

Reports & Pages

Show the live queue to staff and the administrator.

Advantages & Future Enhancements

Advantages

- Reduced waiting time.
- Improved customer experience.
- Less manual work.
- Better counter management.
- Accurate digital records.

Future enhancements

- SMS and WhatsApp notifications.
- QR code token generation.
- Mobile application.
- Estimated waiting time.
- Advanced analytics.

Conclusion

- A digital queue system built on Zoho Creator automates registration, token generation and queue tracking.
- Zoho Creator's low-code tools make the system quick to build and easy to maintain.
- Automation improves counter management and overall service efficiency.

Thank You

Questions & Answers

